

Learnable optical model for hologram correction

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Introduction to Holography

➤ Background

- *Holography*
 - Technique where wavefronts can be recorded then reconstructed
- *Computer-generated holography (CGH)*
 - Apply computer algorithms to generate or prepare holograms
- *Metasurface holography*
 - Use of subwavelength nanostructures
 - Design geometry of nanoposts to shape light wavefront for hologram generation [1] [2] [3]

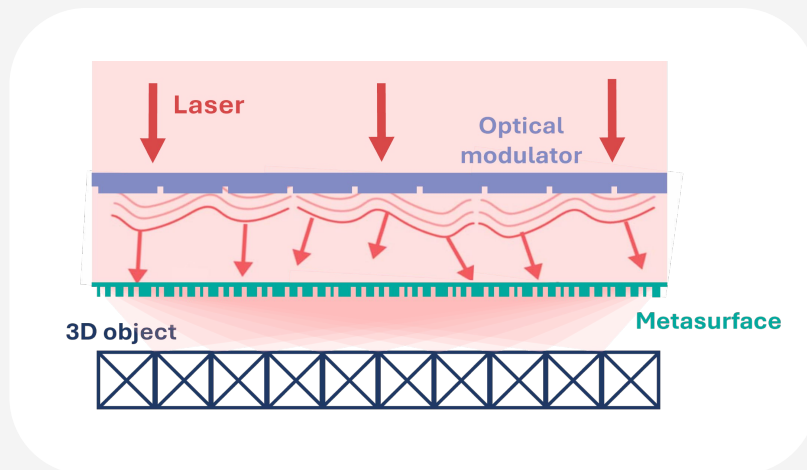
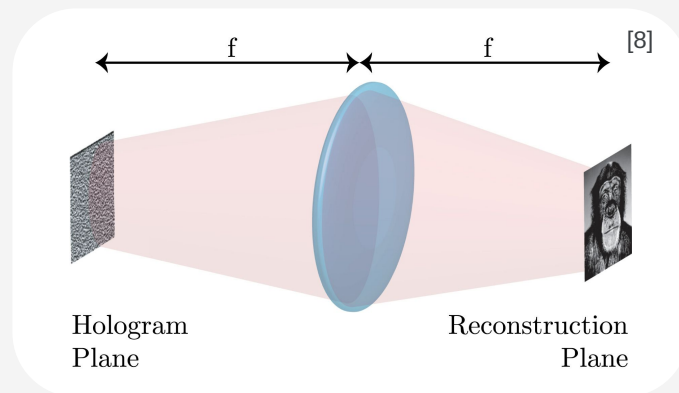
➤ Applications

- AR/VR display, medical imaging, optical tweezers, 3D printing, etc. [4] [5]

➤ Objective

- Hologram quality suffers from experimental and simulation mismatch [6] [7]
 - Imperfections along optical path (wavefront distortion, misalignment, nonideal SLM response, etc.)

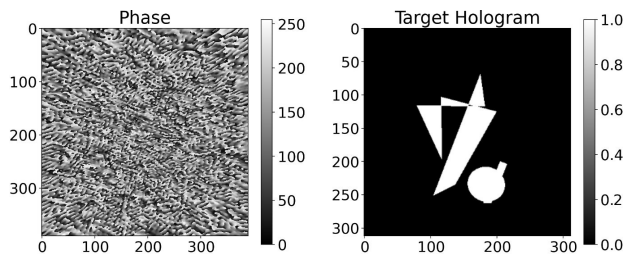
How can we use computational methods to correct for this mismatch?



Methodology

➤ Train Model

- 2k data set
- Shapes
- Borders
- Images

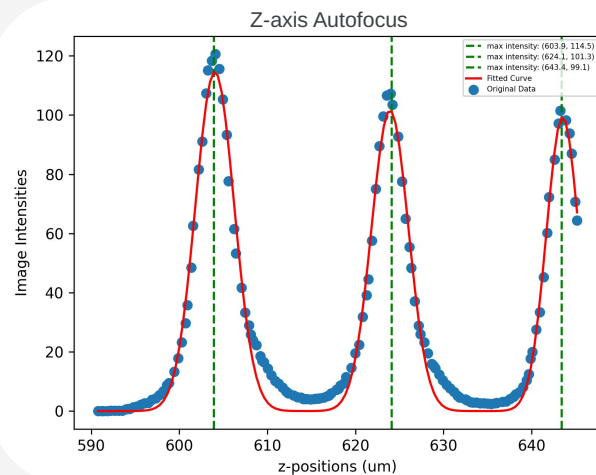
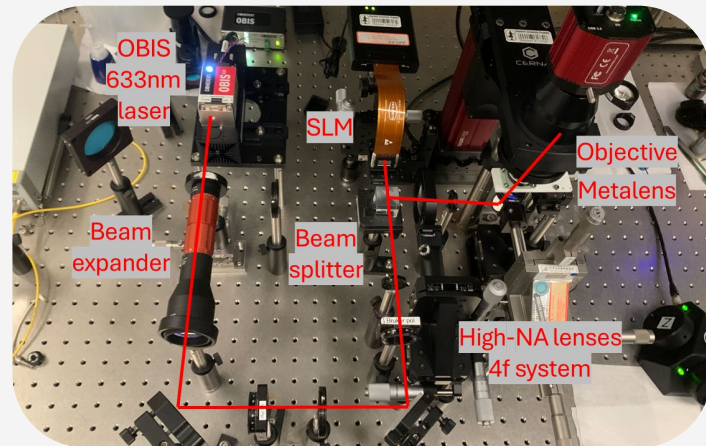
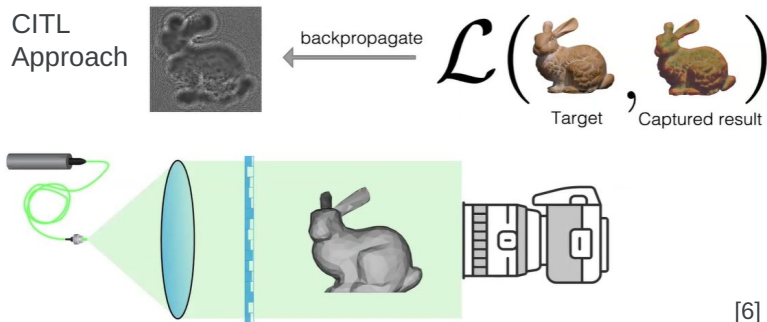


➤ Camera-in-the-loop

- Compare target with intermediate results

➤ Autofocus

- Align XY planes with random dots
- Find ideal Z plane at peak image intensity



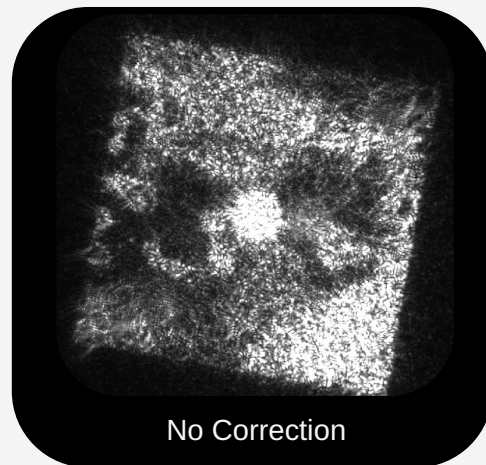
Results & Future Work

➤ Results

- Trained model with CITL and XYZ autofocus performs the best
 - Loss: $2.84e-01$
- Can partially correct for zeroth-order light (non-diffracted light)

➤ Future Work

- Extend model to account for different types of metasurfaces
- Add learning rate optimization for model and CITL
- Train model for 3D holograms



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